

Pranjal Upadhyay

Software Engineer at Amazon

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EDUCATION

University of Texas at Dallas

MS in Computer Science

Related Courses—Machine Learning, Database Design, Big Data Analytics, Artificial Intelligence

August.2021-June.2023

Delhi Technological University, Delhi, India

Bachelor of Technology

Related Courses—Operating Systems, Mechatronics, Operations Research, Programming Fundamentals

August.2015-June.2019

EXPERIENCE

Software Engineer at Amazon (AWS) (Seattle, WA)

August.2023-Present

- Designed, developed and deployed Customer-Log backfill application using AWS Cloudtrail APIs in **Java** to backfill 5 million customer records. Optimised the backfill operation by implementing **multi-threading** to reduce the backfill time by 90 percent.
- Developed ListResourcePolicy API for cross account access using **Java Spring boot** to list out existing IAM policies that dictate the authorization of resources onto a Kinesis Stream
- Architected the Migration of the AWS Kinesis Elastic Load Balancers transitioning (termed as Tardigrade Migration) eliminating virtual IP address, making the scaling and diagnosis of the Load balancers easier.
- Re-architected AWS Kinesis Data Streams Load-balancer scaling architecture eliminating the need to add new host groups in order to scale up the load balancers cutting the cost and time of LB-Scaling by 40 percent. Performed Load-Balancer scaling in Dublin (DUB) and Virginia (IAD) regions.

Software Engineer at Copart (Dallas, Texas)

September.2022-June.2023

- Worked on web-based applications and services using Java, Spring, Hibernate, AngularJS, and JavaScript, collaborating with senior engineers and management to optimize application design of Copart's document delivery service and auction software.
- Optimised and improved API performance for Copart's Document delivery service API in Spring Boot for image upload using Redis Caching.
- Did UI improvements for document upload dashboard in AngularJS

Software Engineer at Bluekactus (New Delhi, India)

July.2019-July.2021

- Developed a [Costing Engine framework](#) using Deep Learning and NLP, achieving 70 percent confidence. eliminating the need of garment merchandisers, thus saving per annum company cost by more than 10 percent.
- Envisioned and developed [ZeDQ mobile application](#) using React Native, serving 500+ users. Developed an Auto-form application to generate forms for the application to make the development process quicker.
- Empowered clients' purchase order making capability by developing a RASA-powered Virtual AI assistant for clients, streamlining tasks and saving 60
- Integrated Google Firebase push notification service with Azure server enabling periodic polling of SQL server database using NodeJS and Cron Job, sending out notifications to Android and IOS devices in intervals of 15 minutes.

TECHNICAL SKILLS

- Programming Languages** Javascript, Python, Spark, Java, SQL, Bash
- Technologies** AWS Kinesis, Node.js, React Native, Kafka, Pyspark MLlib, Flask, Numpy, Pandas, Scikit-learn, PySpark, Spring Boot, Kafka
- Databases** MongoDB, MariaDB, MySQL, SQLite, Impala, Hive, Elasticsearch, Cassandra
- Operating Systems** Linux, Ubuntu, Kali Linux

PROJECTS

Tweet Processing and Classification Using PySpark MLlib Pipelines [\[Databricks Notebook link\]](#)

2022

- Synthesized a PySpark Pipeline for executing Stopword Removal, Tokenization, Term Hashing, Label Indexing and finally classification of [US airport tweets](#) as Positive, Negative or Neutral using Logistic Regression, achieving more than 70 percent accuracy and F1 score.

Apache Kafka Twitter Stream Sentiment Analysis using ELK stack [\[Github link\]](#)

2022

- Contrived and capacitated real-time [analysis](#) of more that 250 tweets per minute using Kafka, Elasticsearch, Logstash and Kibana and Twitter's tweepy API

Community detection in Orkut Network using Girvan Newman algorithm [\[Code link\]](#) [\[Paper link\]](#)

2022

- Brainstormed and effectuated a novel implementation of the Girvan Newman Algorithm detecting communities in Orkut Network using data of nearly 1000 users sampled from a dataset of a million users, coded out using PySpark's Graphframes framework creating a potentially new library.